

Data Flow Diagram

Date	30 September 2026
Team ID	SWTID-2026-7960
Project Name	Import Data using Transform Maps (Spreadsheet) in ServiceNow
Maximum Marks	2 Marks

Data Flow Diagram Components & Description (ServiceNow Transform Maps Project):

1. External Entities (Sources / Users):

- **HR Administrator / Data Provider:**
 - **Role:** Sends or uploads raw employee spreadsheet files (.csv / .xlsx) into the ServiceNow instance.
- **System Auditor / Manager:**
 - **Role:** Receives final imported employee reports, dashboards, and transaction status summaries.

2. Processes (Transform Operations):

- **Process 1.0 (Load & Stage Data):**
 - **Description:** Receives the uploaded spreadsheet file from the HR Administrator and loads the raw rows into the staging table (u_employee_import).
- **Process 2.0 (Map Fields & Apply Coalesce):**
 - **Description:** Processes the mapping configuration using Transform Maps, evaluates the Coalesce rule on unique identifiers (e.g., Employee ID) to check for existing records, and handles data translation.
- **Process 3.0 (Insert / Update Target Records):**
 - **Description:** Transforms and writes the validated data into the final target table (u_employee_test / employee user table) while updating the transaction logs.

3. Data Stores (Tables & Logs):

- **Data Store D1 (Staging Table - u_employee_import):**
 - **Description:** A temporary holding table where raw spreadsheet data resides

before transformation.

- **Data Store D2 (Target Table - Employee Master Table):**
 - *Description:* The official repository where clean, validated employee records are stored permanently.
- **Data Store D3 (Transform History / System Logs):**
 - *Description:* Stores execution logs detailing whether records were inserted, updated, or skipped.

4. Data Flows (Movement of Data):

- **Flow 1: HR Administrator → Process 1.0 (Uploads Employee Spreadsheet File)**
- **Flow 2: Process 1.0 → Data Store D1 (Stores Raw Data in Staging Table)**
- **Flow 3: Data Store D1 → Process 2.0 (Reads Staged Data for Mapping & Matching)**
- **Flow 4: Process 2.0 → Data Store D3 (Writes Transaction State & Status History)**
- **Flow 5: Process 2.0 → Process 3.0 (Passes Mapped & Coalesce-checked Data)**
- **Flow 6: Process 3.0 → Data Store D2 (Saves Final Inserted/Updated Records into Target Table)**
- **Flow 7: Data Store D3 → System Auditor (Generates Import Summary Reports & Audit Logs)**

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example

